

Can quality-of-life indicators distinguish between countries with different living standards?

Understanding Quality of Life in the European Union: A Linear Discriminant Analysis

supervising professor: Obretin Alexandru-Marius

student: Smeu Estera-Valentina

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Abstract

This study explores whether quality-of-life indicators can effectively distinguish between EU countries with different living standards. Data was obtained from the World Bank Open Data platform (<https://data.worldbank.org/>) and include measures of health, education, and access to essential services for a cross-section of countries. After creating quality-of-life groups based on a composite score of the indicators and standardizing the data, I applied an Linear Discriminant Analysis (LDA) to identify combinations of indicators that best separate these groups. The results show that health expenditure, life expectancy, and school enrollment are the strongest discriminators of quality-of-life differences among EU countries. Visualization of the first two discriminant functions reveals a clear separation between low and high-quality-of-life countries, while middle groups show partial overlap. Overall, the findings suggest that a small number of human-centered indicators can summarize complex development differences and provide meaningful insights into quality-of-life patterns within the European Union.

Introduction

Quality of life is a central concern in development studies - going beyond income to include health, education, and basic living conditions. While economic indicators are important, they do not fully capture how people experience everyday life.

The World Bank provides a wide range of comparable indicators that allow researchers to examine quality-of-life differences across countries. The Stiglitz-Sen-Fitoussi Commission emphasized that *“what we measure affects what we do”* and that societies must look beyond GDP to understand well-being (Stiglitz, Sen & Fitoussi, 2009). Understanding whether these indicators align with broad quality-of-life categories can help policymakers and researchers better interpret development gaps, even within relatively homogeneous regions like the European Union.

Can quality-of-life indicators distinguish between countries with different living standards?

To address the question, this study uses Linear Discriminant Analysis, a supervised method designed to find linear combinations of variables that best separate predefined groups. Fisher, who introduced the method, described its purpose as *“finding the linear combination of variables which best discriminates between groups”*. Given that all EU countries are classified as high-income, we create quality-of-life groups based on a composite score of multiple indicators to examine nuanced differences within the region.

Data description

Data Source:

World Bank Open Data - <https://data.worldbank.org/>

Units of Observation:

European Union Member Countries

Time Period:

Most recent year with sufficient data availability (around 2021)

Variables Used:

<i>Variable</i>	<i>Description</i>
<i>LIFE_EXP</i>	Life expectancy at birth, total (years)
<i>INF_MORT</i>	Mortality rate, infant (per 1,000 live births)
<i>SCHOOL</i>	School enrollment, secondary (% gross)
<i>ELECTRIC</i>	Access to electricity (% of population)
<i>HEALTH_EXP</i>	Current health expenditure (% of GDP)
<i>QoL_GROUP</i>	Create based on composite score

Data processing:

Since all EU countries are classified as high-income, I used a composite score approach to find the quality-of-life groups:

1. Standardized all quality-of-life indicators using z-scores;
2. Inverted INF_MORT, so higher values indicate better outcomes;
3. Calculated mean composite score across all indicators;
4. Divided countries into three groups (High, Medium, Low) using tertiles.

Methodology

Linear Discriminant Analysis identifies weighted combinations of quality of life indicators that maximize differences between predefined quality-of-life groups while minimizing variation within each group. In simple terms, it finds the dimensions along which EU countries differ the most in terms of daily living conditions.

The method assumes linear relationships, approximate normality within groups, and similar variability across groups. Standardization is essential because the indicators are measured in different units.

LDA is appropriate here because the research question focuses on distinguishing known groups rather than discovering unknown patterns.

Results

The analysis successfully separated EU countries into three quality-of-life groups:

Country	QoL_GROUP	QoL_SCORE	LIFE_EXP	INF_MORT	SCHOOL	ELECTRIC	HEALTH_EXP	LD1	LD2
Sweden	High	1,076321	82,7	2,2	105	100	10,9	-4,50496	-0,91321
Finland	High	0,855614	81,7	2,3	103	100	10,7	-3,39592	-0,80992
Netherlands	High	0,746934	81,9	2,9	103	100	10,5	-3,01525	-1,32139
Denmark	High	0,692521	81,2	2,8	102	100	10,8	-2,66881	-1,44886
Germany	High	0,691148	81,3	3,2	102	100	11,3	-2,77431	-2,20551
France	High	0,67436	82,5	3,5	101	100	11,2	-3,16349	-1,46979
Austria	High	0,570725	81,8	3,3	101	100	10,5	-2,417	-1,1917
Spain	High	0,568211	83,8	2,7	99	100	9,1	-2,99421	2,303382
Ireland	High	0,510494	82,1	2,9	104	100	7,9	-1,7178	0,076618
Luxembourg	Medium	0,497799	82,2	2,7	105	100	7	-1,50807	0,604015
Belgium	Medium	0,444613	81,5	3,4	100	100	10,2	-1,83979	-0,96036
Malta	Medium	0,428289	82,5	2,5	99	100	8,5	-1,91737	2,166606
Italy	Medium	0,32567	83,1	3,3	98	100	8,8	-1,83123	1,671508
Portugal	Medium	0,215356	81	2,8	97	100	9	-0,77382	1,287167
Slovenia	Medium	0,149604	81	2,5	97	100	8	-0,36754	2,289592
Cyprus	Medium	0,115449	81	3	98	100	8	-0,17851	1,315721
Greece	Medium	0,030974	81	3,5	98	100	8	0,146931	0,687389
Czech Republic	Medium	-0,002	79,3	2,5	99	100	7	1,067609	1,225693
Estonia	Low	-0,1924	78,9	3,3	99	100	6,8	1,968952	0,110566
Croatia	Low	-0,40832	78,5	4	97	100	7,1	2,77725	-0,50053
Slovakia	Low	-0,54788	77,2	3,5	96	100	6,5	3,795229	0,085233
Hungary	Low	-0,57445	76,9	3,7	95	100	7,2	3,835979	-0,43452
Poland	Low	-0,64604	78,5	4,1	94	100	6,5	3,612488	0,785677
Lithuania	Low	-0,65083	76,4	3,8	96	100	6,6	4,451003	-0,82407
Latvia	Low	-0,76963	75,9	3,9	95	100	6,5	5,038069	-0,83526
Bulgaria	Low	-2,08139	75,6	6,5	91	99	7,5	2,79288	-0,96649
Romania	Low	-2,72114	75	7,8	88	99	5,6	5,58168	-0,72755

Table 1

- High QoL: Austria, Denmark, Finland, France, Germany, Ireland, Netherlands, Spain, Sweden

- Medium QoL: Belgium, Cyprus, Czech Republic, Greece, Italy, Luxembourg, Malta, Portugal, Slovenia

- Low QoL: Bulgaria, Croatia, Estonia, Hungary, Latvia, Lithuania, Poland, Romania, Slovakia

The first discriminant function (LD1) explains 92.0% of the variance, with health expenditure (coefficient: 3.89), life expectancy (3.69), and school enrollment (2.67) being the strongest positive contributors.

Variable	LD1	LD2
LIFE_EXP	3,685036	-6,66934
INF_MORT	-1,12788	3,444839
SCHOOL	2,670666	-1,5347
ELECTRIC	-2,64264	4,261995
HEALTH_EXP	3,892984	-3,49186

Table 2

The second discriminant function (LD2) explains 8.0% of the variance. Visualization of the discriminant space through scatter plots reveals clear separation between groups (Figure 1), while correlation analysis shows relationships among the quality-of-life indicators (Figure 2).

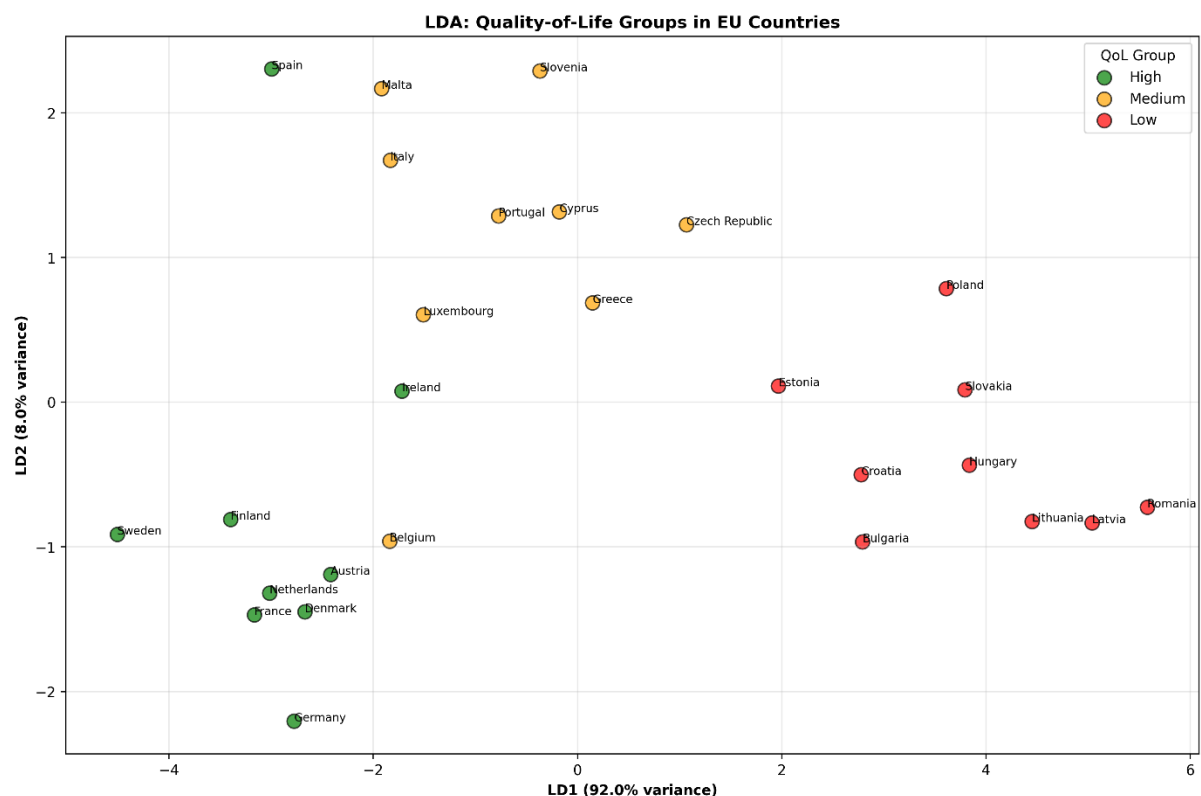


Figure 1

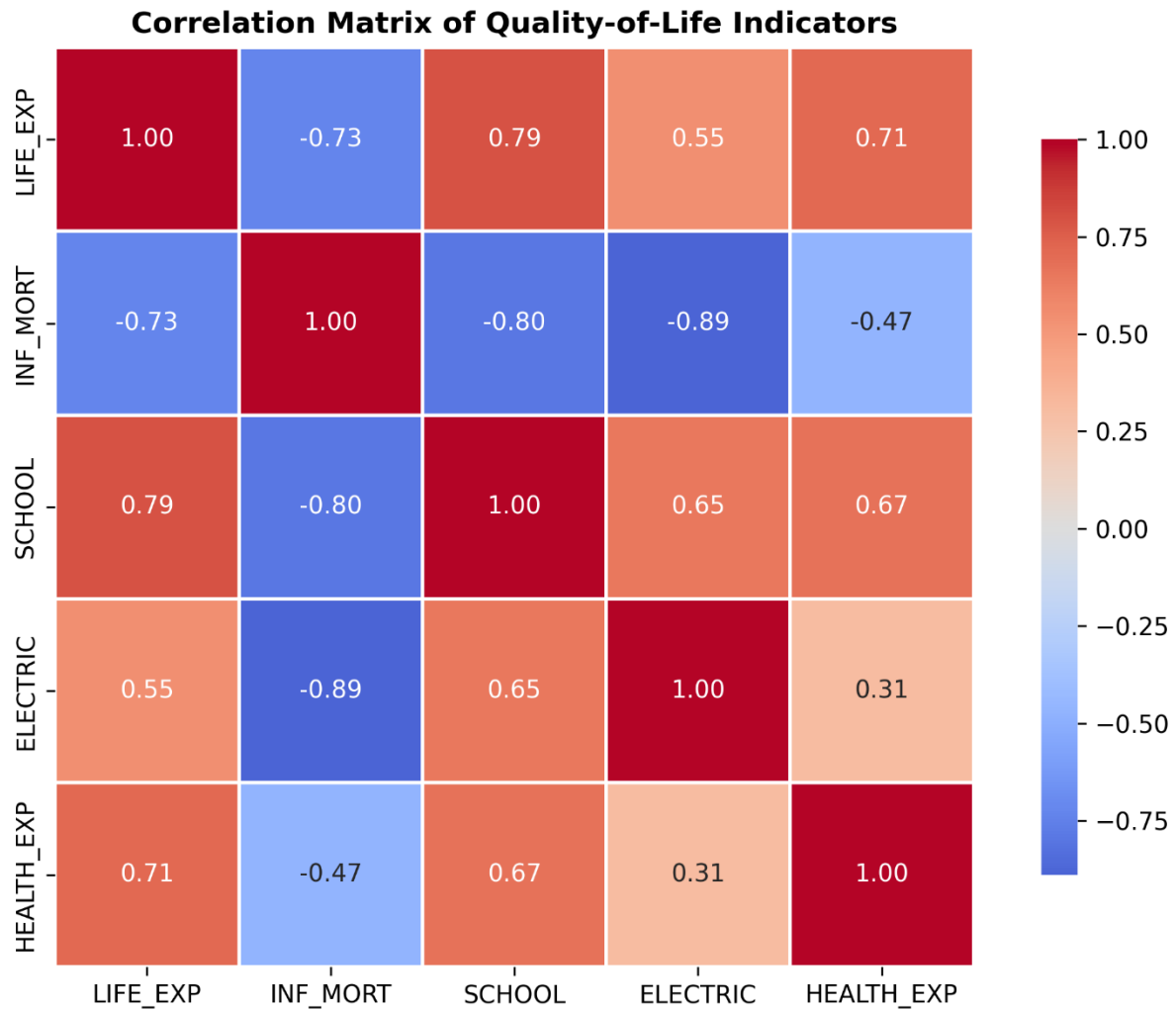


Figure 2

Interpretation

- LD1 reflects overall quality of life (health and education indicators) (Table 2 and Figure 1);
- Health expenditure and life expectancy contribute positively (Table 2);
- Infant mortality contributes strongly in the opposite direction (Table 2);
- Medium quality-of-life countries overlap, indicating gradual transitions within the EU (Figure 1).

Discussion

The results suggest that quality-of-life indicators form coherent dimensions that align well with quality-of-life differences within the European Union. Health expenditure and education are especially important in separating countries at different

development stages, even within a region where all countries are classified as high-income.

A key strength of this analysis is its intuitive interpretation and focused approach, making it suitable for communication with non-technical audiences. The use of a streamlined methodology with clear visualizations and summary tables facilitates understanding of the results. However, the creation of quality-of-life groups based on the data itself is a limitation, as it may reflect statistical patterns rather than external validation criteria.

Future research could incorporate subjective well-being indicators or analyze changes over time.

Conclusion

This study shows that everyday quality-of-life indicators can meaningfully distinguish EU countries with different quality-of-life levels, even within a relatively homogeneous high-income region. Linear Discriminant Analysis provides a clear and interpretable framework for summarizing development differences in a human-centered way, revealing that health and education indicators are particularly important discriminators within the European Union.

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